

AIAC Assignment-14.5

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Batch :24

Task 1 – Image Gallery Website

Create an image gallery web page.

Requirements:

- Display images in a grid layout.
- Use AI to:
 - o Optimize layout using CSS Grid.
 - o Add hover zoom or overlay effects.
 - o Ensure mobile responsiveness.

Code:

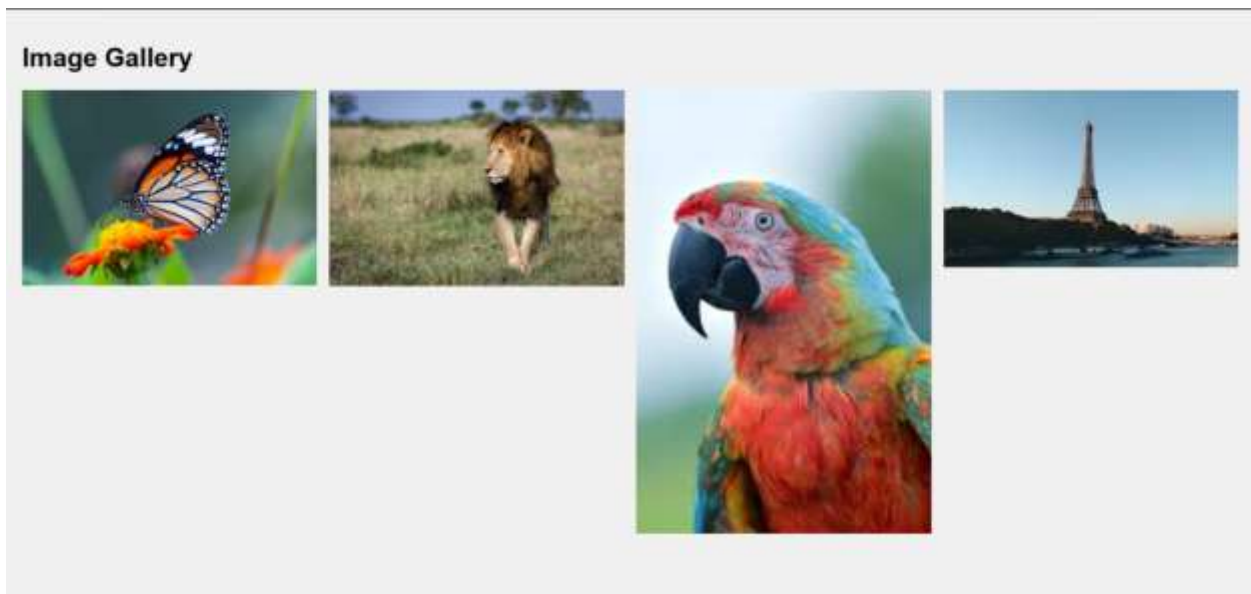
```
1 <!--Create an image gallery web page.
2 Display images in a grid layout.
3 Optimize layout using CSS Grid.
4 Add hover zoom or overlay effects.
5 Ensure mobile responsiveness.-->
6 <!DOCTYPE html>
7 <html lang="en">
8 <head>
9   <meta charset="UTF-8">
10  <meta name="viewport" content="width=device-width, initial-scale=1.0">
11  <title>Image Gallery</title>
12  <style>
13    body {
14      font-family: Arial, sans-serif;
15      margin: 0;
16      padding: 20px;
17      background-color: #f0f0f0;
18    }
19    .gallery {
20      display: grid;
21      grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
22      gap: 15px;
23    }
24    .gallery-item {
25      position: relative;
26      overflow: hidden;
27    }
28    .gallery-item img {
29      width: 100%;
30      height: auto;
31      transition: transform 0.3s ease;
32    }
33    .gallery-item:hover img {
34      transform: scale(1.1);
35    }
36    @media (max-width: 600px) {
37      .gallery {
38        grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
39      }
40    }
41  </style>
42 </head>
43 <body>
```

```

44     <h1>Image Gallery</h1>
45     <div class="gallery">
46         <div class="gallery-item">
47             
48         </div>
49         <div class="gallery-item">
50             
51         </div>
52         <div class="gallery-item">
53             
54         </div>
55         <div class="gallery-item">
56             
57         </div>
58         <!-- Add more images as needed -->
59     </div>
60 </body>
61 </html>

```

Output:



Task 2 – Profile Card Generator

Create a profile card web page.

Requirements:

- Display user image, name, role, and social media links.
- Use AI to:
 - o Suggest modern UI colors and fonts.

- o Design the card using Flexbox.
- o Add hover animation and shadow effects

Code:

```

C:\Users\PC> PC > Downloads > src > profileCardGenerator.html > html > head > style > .profile-cardhover
1  <!--Create a profile card web page.
2  Display user image, name, role, and social media links.
3  Suggest modern UI colors and fonts.
4  Design the card using Flexbox.
5  Add hover animation and shadow effects-->
6  <!DOCTYPE html>
7  <html lang="en">
8  <head>
9      <meta charset="UTF-8">
10     <meta name="viewport" content="width=device-width, initial-scale=1.0">
11     <title>Profile Card</title>
12     <style>
13         body {
14             font-family: 'Arial', sans-serif;
15             background-color: #2f2fac;
16             display: flex;
17             justify-content: center;
18             align-items: center;
19             height: 100vh;
20             margin: 0;
21         }
22         .profile-card {
23             background-color: #fefeb0;
24             border-radius: 10px;
25             box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
26             width: 300px;
27             text-align: center;
28             padding: 20px;
29             transition: transform 0.3s ease, box-shadow 0.3s ease;
30         }
31         .profile-card:hover {
32             transform: translateY(-10px);
33             box-shadow: 0 8px 16px rgba(0, 0, 0, 0.2);
34         }
35         .profile-image {
36             width: 100px;
37             height: 100px;
38             border-radius: 50%;
39             object-fit: cover;
40             margin-bottom: 15px;
41         }
42         .profile-name {
43             font-size: 1.5em;
44             margin-bottom: 5px;
45         }
46         .profile-role {
47             color: #0e0b0b;
48             margin-bottom: 15px;
49         }
50         .social-links a {
51             color: #100e0e;
52             text-decoration: none;
53             margin: 0 10px;
54             font-size: 1.2em;
55         }
56         .social-links a:hover {
57             color: #8080;
58         }
59     </style>
60 </head>
61 <body>
62     <div class="profile-card">
63         
64         <h2 class="profile-name">John Doe</h2>
65         <p class="profile-role">Software Engineer</p>
66         <div class="social-links">
67             <a href="#" target="_blank">LinkedIn</a>
68             <a href="#" target="_blank">Twitter</a>
69             <a href="#" target="_blank">GitHub</a>
70         </div>
71     </div>
72 </body>
73 </html>

```

Output:



Task 3 -Develop a dynamic shopping cart system.

Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
 - o Manage cart state using JavaScript objects.
 - o Persist cart data in localStorage.

Code:

```
C:\Users> PC > Downloads > aac > DynamicShoppingCart.html > @html
1 <!-- Develop a dynamic shopping cart system.
2 Add, remove, and update product quantities.
3 Display real-time total price.
4 Manage cart state using JavaScript objects.
5 Persist cart data in localStorage.
6 -->
7 <!DOCTYPE html>
8 <html lang="en">
9 <head>
10   <meta charset="UTF-8">
11   <meta name="viewport" content="width=device-width, initial-scale=1.0">
12   <title>Dynamic Shopping Cart</title>
13   <style>
14     body {
15       font-family: Arial, sans-serif;
16       margin: 20px;
17     }
18     .product {
19       display: flex;
20       justify-content: space-between;
21       margin-bottom: 10px;
22     }
23     .cart {
24       margin-top: 20px;
25     }
26     .cart-item {
27       display: flex;
28       justify-content: space-between;
29       margin-bottom: 10px;
30     }
31     .total {
32       font-weight: bold;
33     }
34   </style>
35 </head>
36 <body>
37   <h1>Dynamic Shopping Cart</h1>
38   <div id="products">
39     <div class="product">
40       <span>Product 1 - $10</span>
41       <button onclick="addToCart('Product 1', 10)">Add to Cart</button>
42     </div>
43     <div class="product">
44       <span>Product 2 - $20</span>
45       <button onclick="addToCart('Product 2', 20)">Add to Cart</button>
```

```
46     </div>
47     <div class="product">
48       <span>Product 3 - $50</span>
49       <button onclick="addToCart('Product 3', 30)">Add to Cart</button>
50     </div>
51   </div>
52
53   <div class="cart">
54     <h2>Shopping Cart</h2>
55     <div id="cart-items"></div>
56     <div class="total">Total: <span id="total-price">0</span></div>
57   </div>
58
59   <script>
60     let cart = JSON.parse(localStorage.getItem('cart')) || [];
61
62     function addToCart(product, price) {
63       if (cart[product]) {
64         cart[product].quantity += 1;
65       } else {
66         cart[product] = { price: price, quantity: 1 };
67       }
68       updateCart();
69     }
70
71     function removeFromCart(product) {
72       delete cart[product];
73       updateCart();
74     }
75
76     function updateCart() {
77       const cartItems = document.getElementById('cart-items');
78       cartItems.innerHTML = '';
79       let totalPrice = 0;
80
81       for (let product in cart) {
82         const item = cart[product];
83         totalPrice += item.price * item.quantity;
84         cartItems.innerHTML += `
85         <div class="cart-item">
86           <span>${product} - ${item.price} x ${item.quantity}</span>
87           <button onclick="removeFromCart('${product}')">Remove</button>
```

```

88         </div>
89     };
90 }
91
92     document.getElementById('total-price').textContent = totalPrice.toFixed(2);
93     localStorage.setItem('cart', JSON.stringify(cart));
94 }
95
96     updateCart();
97 </script>
98 </body>
99 </html>

```

Output:



Task 4 – Role-Based Access Control Application

Create a role-based web application.

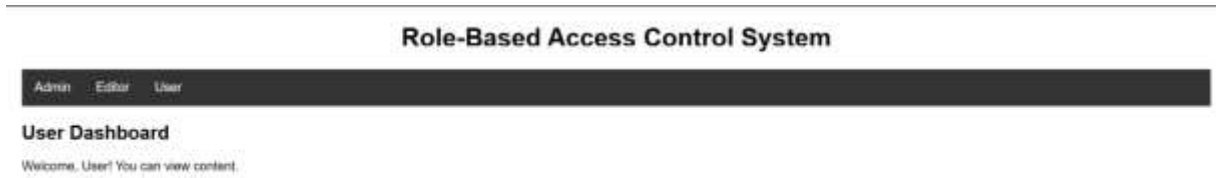
Requirements:

- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
 - o Design access control logic.
 - o Maintain secure state transitions.
 - o Create scalable UI components.

Code:

```
C:\Users> PC > Downloads > mac > roleBasedAccessControl.html > html > body > navbar > a
1  <!-- Role-Based Access Control System
2  | Create a role-based web application.
3  | Admin, Editor, User roles.
4  | Conditional UI rendering.
5
6  | Design access control logic.
7  | Maintain secure state transitions.
8  | Create scalable UI components.
9  -->
10 <!DOCTYPE html>
11 <html lang="en">
12 <head>
13   <meta charset="UTF-8">
14   <meta name="viewport" content="width=device-width, initial-scale=1.0">
15   <title>Role-Based Access Control</title>
16   <style>
17     body {
18       font-family: Arial, sans-serif;
19       margin: 20px;
20     }
21     .navbar {
22       background-color: #333;
23       overflow: hidden;
24     }
25     .navbar a {
26       float: left;
27       display: block;
28       color: white;
29       text-align: center;
30       padding: 14px 16px;
31       text-decoration: none;
32     }
33     .navbar a:hover {
34       background-color: #ddd;
35       color: black;
36     }
37     .content {
38       margin-top: 20px;
39     }
40   </style>
41 </head>
42 <body>
43   <h1 style="text-align: center;">Role-Based Access Control System</h1>
44   <div class="navbar">
45     <a href="#" onclick="setRole('Admin')">Admin</a>
46     <a href="#" onclick="setRole('Editor')">Editor</a>
47     <a href="#" onclick="setRole('User')">User</a>
48   </div>
49
50   <div class="content" id="content">
51     <!-- Content will be rendered based on role -->
52   </div>
53
54   <script>
55     let currentRole = 'User'; // Default role
56
57     function setRole(role) {
58       currentRole = role;
59       updateContent();
60     }
61
62     function updateContent() {
63       const contentDiv = document.getElementById('content');
64       contentDiv.innerHTML = ''; // Clear existing content
65
66       if (currentRole === 'Admin') {
67         contentDiv.innerHTML = '<h2>Admin Dashboard</h2><p>Welcome, Admin! You have full access.</p>';
68       } else if (currentRole === 'Editor') {
69         contentDiv.innerHTML = '<h2>Editor Dashboard</h2><p>Welcome, Editor! You can edit content.</p>';
70       } else if (currentRole === 'User') {
71         contentDiv.innerHTML = '<h2>User Dashboard</h2><p>Welcome, User! You can view content.</p>';
72       }
73     }
74
75     // Initialize content based on default role
76     updateContent();
77   </script>
78 </body>
79 </html>
```

Output:



Task 5 – Multi-Language Website

Design a multilingual website.

Requirements:

- Switch languages dynamically.
- Persist language preference.
- Use AI to:
 - o Structure translation files.
 - o Handle RTL/LTR layouts.
 - o Improve accessibility.

Code:

```
C:\Users\> PC > Downloads > mac > multilingualWebsite.html > @ @ @ body
1 <!--
2 Design a multilingual website.
3 Switch languages dynamically.
4 Persist language preference.
5
6 Structure translation files.
7 Handle RTL/LTR layouts.
8 Improve accessibility.
9 -->
10
11 <!DOCTYPE html>
12 <html lang="en">
13 <head>
14   <meta charset="UTF-8">
15   <meta name="viewport" content="width=device-width, initial-scale=1.0">
16   <title>Multilingual Website</title>
17   <style>
18     body {
19       font-family: Arial, sans-serif;
20       margin: 20px;
21     }
22     .language-switcher {
23       margin-bottom: 20px;
24     }
25     .content {
26       font-size: 1.2em;
27     }
28     .background {
29       background-color: #f0f0f0;
30       padding: 20px;
31       border-radius: 10px;
32     }
33   </style>
34 </head>
35 <body>
36   <div style="text-align: center;">Multilingual Website</div>
37   <div class="language-switcher">
38     <button onclick="setLanguage('en')">English</button>
39     <button onclick="setLanguage('es')">Spanish</button>
40     <button onclick="setLanguage('fr')">French</button>
41   </div>
42   <div class="content" id="content">
43     <!-- Content will be rendered based on language -->
44   </div>
45
```

```
46 <script>
47   const translations = {
48     en: {
49       welcome: "Welcome to our multilingual website!",
50       description: "This website supports multiple languages. Please select your preferred language."
51     },
52     es: {
53       welcome: "¡Bienvenido a nuestro sitio web multilingüe!",
54       description: "Este sitio web admite varios idiomas. Por favor, selecciona su idioma preferido."
55     },
56     fr: {
57       welcome: "Bienvenue sur notre site web multilingue!",
58       description: "Ce site web prend en charge plusieurs langues. Veuillez sélectionner votre langue préférée."
59     }
60   };
61
62   let currentLanguage = 'en'; // Default language
63
64   function setLanguage(lang) {
65     currentLanguage = lang;
66     localStorage.setItem('preferredLanguage', lang);
67     updateContent();
68   }
69
70   function updateContent() {
71     const content = translations[currentLanguage];
72     document.getElementById('content').innerHTML = `
73       <h2>${content.welcome}</h2>
74       <p>${content.description}</p>
75     `;
76   }
77
78   // Load preferred language on page load
79   window.onload = function() {
80     const savedLanguage = localStorage.getItem('preferredLanguage');
81     if (savedLanguage) {
82       setLanguage(savedLanguage);
83     } else {
84       updateContent();
85     }
86   };
87 </script>
88 </body>
```

Output:



Task 6 – Online Examination System

Create an online exam interface.

Requirements:

- Timer, questions, and submission.
- Prevent multiple submissions.
- Use AI to:
 - o Manage exam state.
 - o Improve exam UX.
 - o Ensure accessibility.

Code:

```
C:\Users\PC\Desktop> cd C:\onlineexamination\exam.html > cd html > cd body
1  <!--
2  Create an online exam interface.
3
4  Timer, questions, and submission.
5  Prevent multiple submissions.
6  Manage exam state.
7  Improve exam UX.
8  Ensure accessibility.
9  -->
10 <!DOCTYPE html>
11 <html lang="en">
12 <head>
13   <meta charset="UTF-8">
14   <meta name="viewport" content="width=device-width, initial-scale=1.0">
15   <title>Online Examination System</title>
16 </head>
17 <body>
18   <div class="exam-container">
19     <div class="timer">Time Remaining: 10:00</div>
20     <div class="question">
21       <p>1. What is the capital of France?</p>
22       <div class="options">
23         <div class="option"><input type="radio" name="q1" value="A"> A) Berlin</div>
24         <div class="option"><input type="radio" name="q1" value="B"> B) Madrid</div>
25         <div class="option"><input type="radio" name="q1" value="C"> C) Paris</div>
26         <div class="option"><input type="radio" name="q1" value="D"> D) Rome</div>
27       </div>
28     </div>
29     <div class="question">
30       <p>2. What is 2 + 2?</p>
31       <div class="options">
32         <div class="option"><input type="radio" name="q2" value="A"> A) 3</div>
33         <div class="option"><input type="radio" name="q2" value="B"> B) 4</div>
34         <div class="option"><input type="radio" name="q2" value="C"> C) 5</div>
35         <div class="option"><input type="radio" name="q2" value="D"> D) 6</div>
36       </div>
37     </div>
38     <button type="submit">Submit Exam</button>
39   </div>
40 </body>
```

```
46 <div class="exam-container">
47   <div class="timer">Time Remaining: 10:00</div>
48   <div class="question">
49     <p>1. What is the capital of France?</p>
50     <div class="options">
51       <div class="option"><input type="radio" name="q1" value="A"> A) Berlin</div>
52       <div class="option"><input type="radio" name="q1" value="B"> B) Madrid</div>
53       <div class="option"><input type="radio" name="q1" value="C"> C) Paris</div>
54       <div class="option"><input type="radio" name="q1" value="D"> D) Rome</div>
55     </div>
56   </div>
57   <div class="question">
58     <p>2. What is 2 + 2?</p>
59     <div class="options">
60       <div class="option"><input type="radio" name="q2" value="A"> A) 3</div>
61       <div class="option"><input type="radio" name="q2" value="B"> B) 4</div>
62       <div class="option"><input type="radio" name="q2" value="C"> C) 5</div>
63       <div class="option"><input type="radio" name="q2" value="D"> D) 6</div>
64     </div>
65   </div>
66   <button type="submit">Submit Exam</button>
67 </div>
68 <script>
69   let timeRemaining = 600; // 10 minutes in seconds
70   const timerElement = document.getElementById('timer');
71   const examForm = document.getElementById('exam-form');
72   let examSubmitted = false;
73
74   const timerInterval = setInterval(() => {
75     if (timeRemaining > 0) {
76       timeRemaining--;
77       const minutes = Math.floor(timeRemaining / 60);
78       const seconds = timeRemaining % 60;
79       timerElement.textContent = `Time Remaining: ${minutes}:${seconds < 10 ? '0' : ''}${seconds}`;
80     } else {
81       clearInterval(timerInterval);
82       if (!examSubmitted) {
83         alert('Time is up! Your exam will be submitted automatically.');
```

```

100     }, 1000);
101
102     examForm.addEventListener('submit', (e) => {
103         e.preventDefault();
104         if (!examSubmitted) {
105             submitExam();
106         } else {
107             alert('You have already submitted the exam.');

```

Output:

Online Examination

Time Remaining: 9:07

1. What is the capital of France?

☐ A) Berlin
☐ B) Madrid
☐ C) Paris
☐ D) Rome

2. What is 2 + 2?

☐ A) 3
☐ B) 4
☐ C) 5
☐ D) 6

Submit Exam